Wearable EEG and Machine Learning for Real-Time Emotional Valence and Mental Fatigue

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Background

Wearable electroencephalography (EEG) devices have become accessible tools for monitoring brain activity in everyday environments, creating new opportunities for real-time detection of emotional valence and mental fatigue. Because these states are closely linked to frontal-cortex activity, signals from the Fp1 and Fp2 electrodes are especially valuable for lightweight wearable systems, though accuracy is often limited by noise, motion artifacts, and the small number of electrodes available in consumer devices. Prior work shows that simplified processing, such as using the Fp1-Fp2 differential signal and targeted time and frequency-domain features, can improve recognition performance while keeping computational demands low. Building on this, our study evaluates frontal EEG data from the DEAP dataset and applies lightweight feature extraction and machine learning classifiers to compare feature sets, time segments, and frequency ranges. The goal is to determine whether a wearable EEG system with machine learning can accurately and efficiently recognize emotional valence and mental fatigue in real-world, mobile applications.

Objectives

-Evaluate the effectiveness of wearable EEG technology using frontal electrodes to record emotional and mental states in real-world application
-Compare machine-learning models and deep-learning models to improve accuracy
-Investigate how different frequency ranges, feature sets, and time windows affect classification of signals to optimize performance

Methods

Using the DEAP emotional EEG dataset, we processed recordings from 32 participants during standardized affective video stimuli. Signals from the frontal Fp1 and Fp2 electrodes were extracted, filtered, and converted into a differential channel to emphasize valence-related activity. Time and frequency-domain features, including Hjorth parameters, zero-crossings, and power spectral density across multiple bands, were computed in short time windows. Machine learning classifiers were implemented to evaluate subject-dependent valence recognition performance. Feature combinations, time segments, and frequency ranges were systematically compared to optimize accuracy while minimizing electrode and computational requirements for wearable applications.

Results

Using raw EEG data from 1-hour excavation tasks in a real construction environment, all three deep learning models demonstrated high performance in classifying operator mental fatigue, with the Bi-LSTM architecture emerging as the standout performer across all metrics, achieving 99.94% accuracy alongside exceptional precision, recall, specificity, and F1-scores. The Bi-LSTM model exhibited robust handling of on-site noise and variability, confirming the viability of wearable, flexible EEG headsets for capturing reliable signals under challenging conditions. Overall, these results underscore the efficacy of deep learning—particularly Bi-LSTM networks—for automated, real-time mental fatigue detection directly from raw EEG inputs, obviating the need for manual feature engineering and paving the way for enhanced safety monitoring on active construction site.

Figure 1: Senior Authors With Multiple Papers of the Top 100 Most Cited Papers

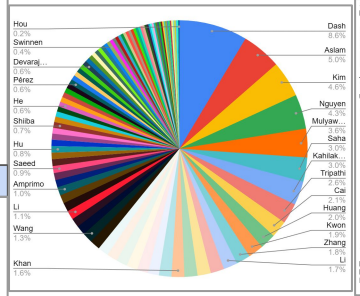


Figure 2: Institutions With Multiple Papers of the Top 100 Most Cited Papers

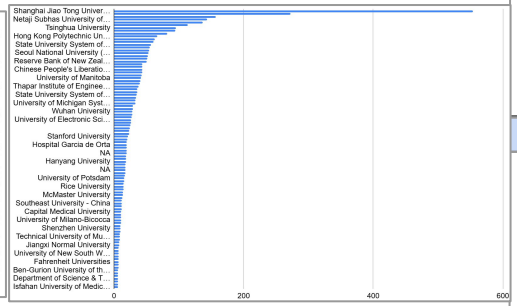


Figure 3: Journals With Multiple Top 100 most cited papers

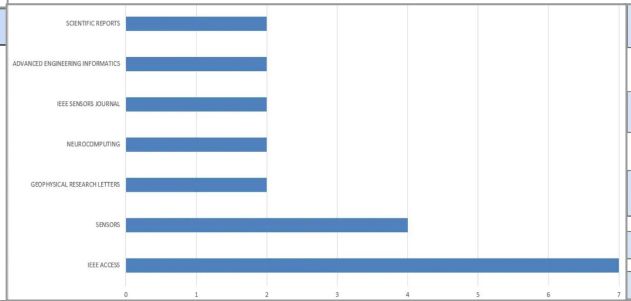
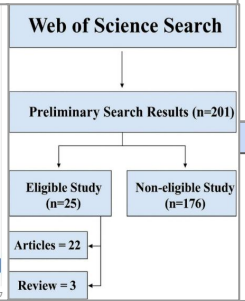


Figure 4: Web of Science Search Method



Discussion

The study results show that lightweight wearable EEG systems using only the frontal Fp1 and Fp2 electrodes are reliable in detecting emotional valence and mental stress when paired with dedicated and efficient machine-learning models. The results, including both the feature-based pipelines and deep learning models, performed strongly, demonstrating that minimal frontal-lobe activity provides enough information to allow accurate real-time monitoring. Bi-LSTM trained models completed with near-perfect accuracy, showing their robustness in noisy environments, where motion and equipment interference are common. While more traditional feature-centered approaches performed well, this underscores the idea that low-power models can support mobile or edge-based EEG devices. Citation trends also highlight rapid growth in EEG-based computing, which only confirms the relevance of this work in modern neuroscience research. The findings from the figures suggest that two-electrode wearable systems with their respective optimized models will provide efficient ways for emotion and fatigue monitoring in real-time.

Conclusion

Lightweight, frontal-electrode wearable EEG devices can accurately track emotional and cognitive states in real-world conditions. We found that both Bi-LSTM deep learning models and traditional feature-based methods deliver strong performance with minimal computing demands, supporting their use in mobile applications. These findings translate to real-world emotional and cognitive monitoring that may be possible outside the lab with low-electrode, high-accuracy systems. The reliability of tracking valence and mental fatigue with frontal electrodes brings in the advancement of future applications in mental health support and human-computer interaction. For future studies, testing whether these models can hold up in diverse environments and populations is important. Going beyond a subject-dependent scenario will significantly improve generalization and robustness in these devices.

Reference

Abdel-Hamid, L. (2023). An efficient machine learning-based emotional valence recognition approach towards wearable EEG. Sensors, 23(3), 1255. https://doi.org/10.3390/s23031255
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